

Expt 4	Design of Low Pass Filter using Windowing Techniques
Date:	4.3. Design and Analysis of an FIR Low-Pass Filter Using the Hanning Window Technique in MATLAB

```

clear;

close all;

%% --- Input Parameters ---

fp = 1000;    % Passband cutoff (Hz)

fs = 4000;    % Sampling frequency (Hz)

N = 50;      % Filter order (even number preferred)

wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- FIR Low Pass using Hanning Window ---

b_hann = fir1(N, wc, 'low', hanning(N+1));

disp('--- Hanning Window Coefficients ---');

disp(b_hann);

%% --- Plot Frequency Responses ---

figure;

freqz(b_hann,1);

title('Low-pass FIR using Hanning Window');

```

